

Results

Study 3 Descriptives Confidence Entry and Exit

Descriptives								
	D1Entry_ArduinoConf	D1Exit_ArduinoConf	D2Exit_ArduinoConf	D1Entry_ProcessingConf	D2Exit_ProcessingConf	D1Entry_CircuitConf	D1Exit_CircuitConf	D2Exit_CircuitCon
N	11	12	10	10	10	10	12	10
Missing	4	3	5	5	5	5	3	5
Median	2	3.50	4.00	1.00	4.00	2.00	3.00	4.00
IQR	2.00	1.00	0.00	1.00	1.00	1.75	1.00	0.750
Minimum	1	3	3	1	3	2	3	2
Maximum	4	4	5	3	5	5	5	4
25th percentile	1.00	3.00	4.00	1.00	3.00	2.00	3.00	3.25
50th percentile	2.00	3.50	4.00	1.00	4.00	2.00	3.00	4.00
75th percentile	3.00	4.00	4.00	2.00	4.00	3.75	4.00	4.00

Frequencies

Frequencies of D1Entry_ArduinoConf			
D1Entry_ArduinoConf	Counts	% of Total	Cumulative %
4	1	9.1%	9.1%
3	3	27.3%	36.4%
2	3	27.3%	63.6%
1	4	36.4%	100.0%

Frequencies of D1Exit_ArduinoConf			
D1Exit_ArduinoConf	Counts	% of Total	Cumulative %
4	6	50.0%	50.0%
3	6	50.0%	100.0%

Frequencies of D2Exit_ArduinoConf			
D2Exit_ArduinoConf	Counts	% of Total	Cumulative %
5	1	10.0%	10.0%
4	7	70.0%	80.0%
3	2	20.0%	100.0%

Frequencies of D1Entry_ProcessingConf			
D1Entry_ProcessingConf	Counts	% of Total	Cumulative %
1	6	60.0%	60.0%
2	3	30.0%	90.0%
3	1	10.0%	100.0%

Frequencies of D2Exit_ProcessingConf			
D2Exit_ProcessingConf	Counts	% of Total	Cumulative %
5	1	10.0%	10.0%
4	5	50.0%	60.0%
3	4	40.0%	100.0%

Frequencies of D1Entry_CircuitConf			
D1Entry_CircuitConf	Counts	% of Total	Cumulative %
2	6	60.0%	60.0%
3	1	10.0%	70.0%
4	2	20.0%	90.0%
5	1	10.0%	100.0%

Frequencies of D1Exit_CircuitConf

D1Exit_CircuitConf	Counts	% of Total	Cumulative %
3	7	58.3%	58.3%
4	3	25.0%	83.3%
5	2	16.7%	100.0%

Frequencies of D2Exit_CircuitConf

D2Exit_CircuitConf	Counts	% of Total	Cumulative %
2	1	10.0%	10.0%
3	2	20.0%	30.0%
4	7	70.0%	100.0%

Frequencies of D1Entry_troubleshootCode

D1Entry_troubleshootCode	Counts	% of Total	Cumulative %
5	1	10.0%	10.0%
4	2	20.0%	30.0%
3	1	10.0%	40.0%
2	1	10.0%	50.0%
1	5	50.0%	100.0%

Frequencies of D1Exit_troubleshootCode

D1Exit_troubleshootCode	Counts	% of Total	Cumulative %
5	2	16.7%	16.7%
4	3	25.0%	41.7%
3	4	33.3%	75.0%
2	3	25.0%	100.0%

Frequencies of D2Exit_troubleshootCode

D2Exit_troubleshootCode	Counts	% of Total	Cumulative %
5	3	30.0%	30.0%
4	2	20.0%	50.0%
3	1	10.0%	60.0%
2	4	40.0%	100.0%

Frequencies of D1Entry_troubleshootCircuit

D1Entry_troubleshootCircuit	Counts	% of Total	Cumulative %
4	3	30.0%	30.0%
2	2	20.0%	50.0%
1	5	50.0%	100.0%

Frequencies of D1Exit_troubleshootCircuit

D1Exit_troubleshootCircuit	Counts	% of Total	Cumulative %
2	3	25.0%	25.0%
3	4	33.3%	58.3%
4	3	25.0%	83.3%
5	2	16.7%	100.0%

Frequencies of D2Exit_troubleshootCircuit

D2Exit_troubleshootCircuit	Counts	% of Total	Cumulative %
2	1	10.0%	10.0%
3	4	40.0%	50.0%
4	3	30.0%	80.0%
5	2	20.0%	100.0%

Study 3 Descriptives Ease of Use

Descriptives		
	D1Exit_EaseShield	D2Exit_EaseSculptingCode
N	10	10
Missing	5	5
Median	4.00	5.00
IQR	1.75	1.00
Minimum	3	3
Maximum	5	5

Frequencies

Frequencies of D1Exit_EaseShield			
D1Exit_EaseShield	Counts	% of Total	Cumulative %
5	4	40.0%	40.0%
4	3	30.0%	70.0%
3	3	30.0%	100.0%

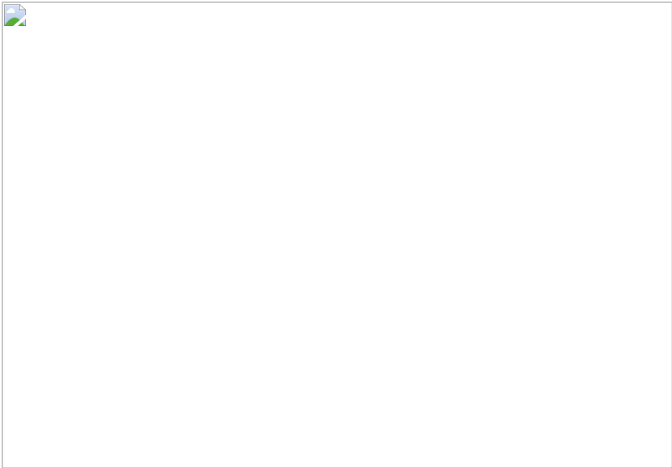
Frequencies of D2Exit_EaseSculptingCode			
D2Exit_EaseSculptingCode	Counts	% of Total	Cumulative %
3	1	10.0%	10.0%
4	3	30.0%	40.0%
5	6	60.0%	100.0%

Study 3 Descriptives Confidence on Entry vs Prior Knowledge Arduino

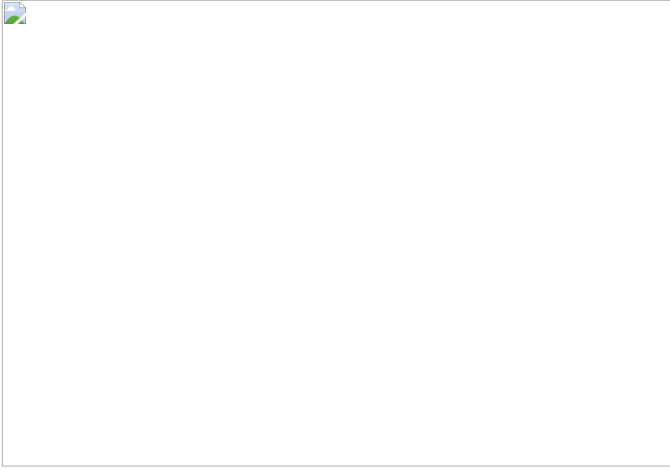
Descriptives						
	PriorPcomp	D1Entry_ArduinoConf	D1Entry_ProcessingConf	D1Entry_CircuitConf	D1Entry_troubleshootCode	D1Entry_troubleshootCircuit
N	yes	5	5	5	5	5
	no	5	5	5	5	5
Missing	yes	0	0	0	0	0
	no	0	0	0	0	0
Median	yes	2	2	2	4	2
	no	1	1	2	1	1
IQR	yes	1.00	1.00	2.00	3.00	3.00
	no	1.00	0.00	1.00	1.00	1.00
Minimum	yes	1	1	2	1	1
	no	1	1	2	1	1
Maximum	yes	4	2	5	5	4
	no	3	3	4	3	4

Plots

D1Entry_ArduinoConf



D1Entry_ProcessingConf



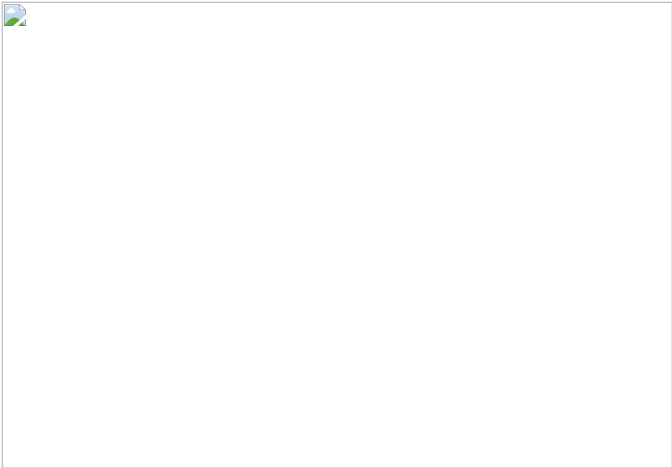
D1Entry_CircuitConf



D1Entry_troubleshootCode



D1Entry_troubleshootCircuit



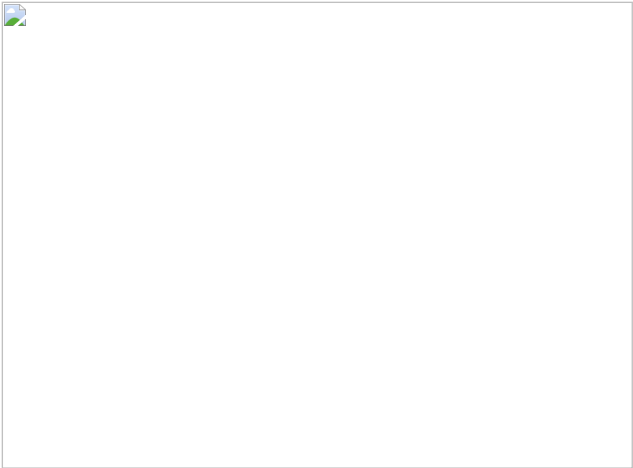
Study 3 Descriptives Confidence on Exit with prior creative code knowledge

Descriptives

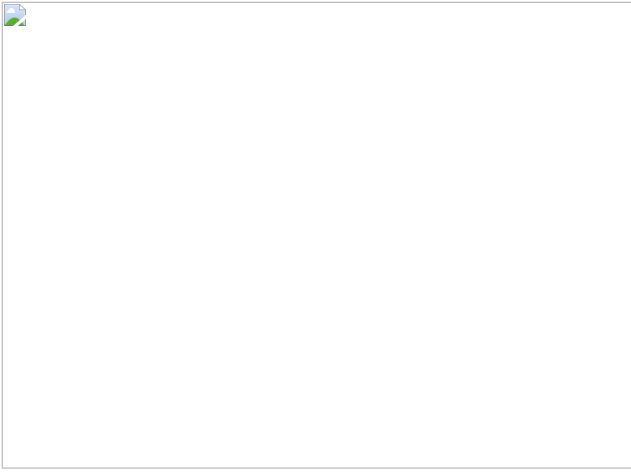
	PriorCC	D2Exit_ArduinoConf	D2Exit_ProcessingConf	D2Exit_CircuitConf	D2Exit_troubleshootCode	D2Exit_troubleshootCircuit
N	no	6	5	5	5	5
	yes	1	1	1	1	1
Missing	no	3	4	4	4	4
	yes	0	0	0	0	0
Median	no	4.00	4	4	3	3
	yes	5	3	4	5	5
IQR	no	0.750	1.00	1.00	2.00	1.00
	yes	0.00	0.00	0.00	0.00	0.00
Minimum	no	3	3	3	2	3
	yes	5	3	4	5	5
Maximum	no	4	4	4	4	4
	yes	5	3	4	5	5

Plots

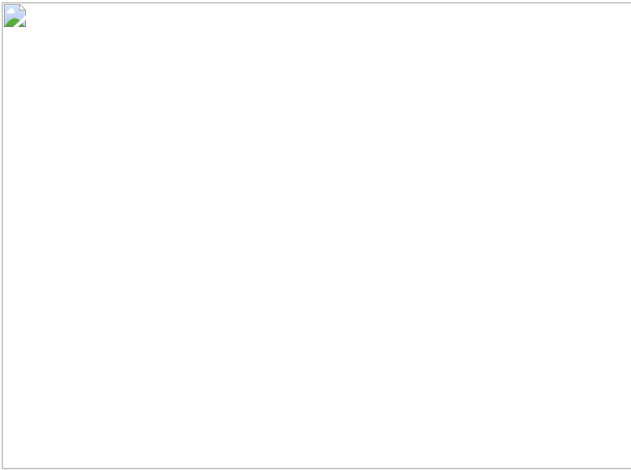
D2Exit_ArduinoConf



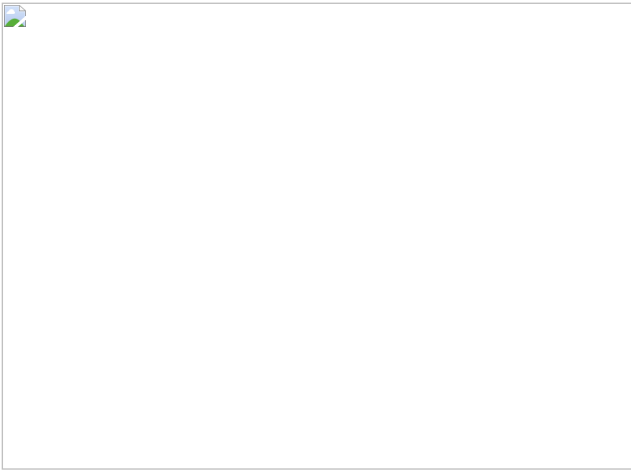
D2Exit_ProcessingConf



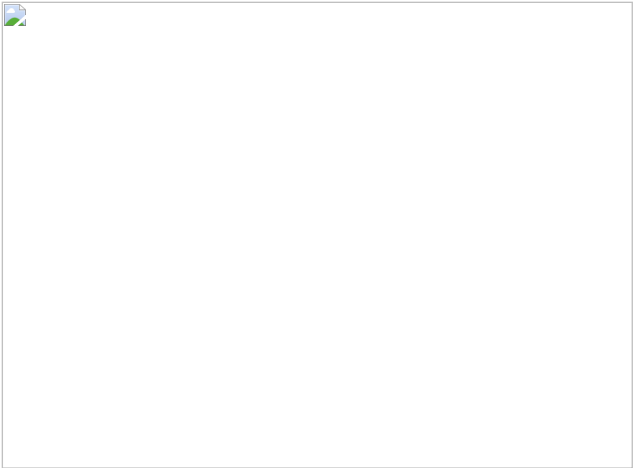
D2Exit_CircuitConf



D2Exit_troubleshootCode



D2Exit_troubleshootCircuit



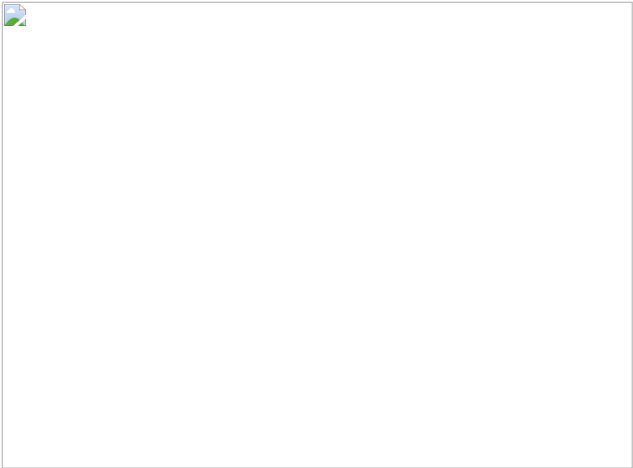
Study 3 Descriptives Day 1 exit confidence with prior Pcomp knowledge

Descriptives

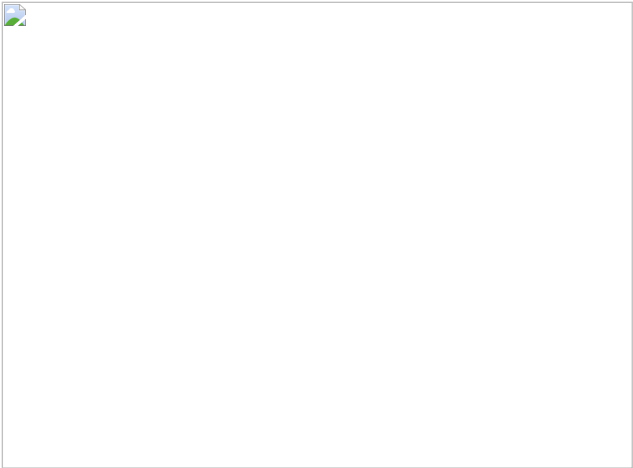
	PriorCC	D1Exit_ArdunoConf	D1Exit_CircuitConf	D1Exit_troubleshootCode	D1Exit_troubleshootCircuit
N	no	9	9	9	9
	yes	1	1	1	1
Missing	no	0	0	0	0
	yes	0	0	0	0
Median	no	3	3	3	3
	yes	4	5	4	5
IQR	no	1.00	1.00	1.00	1.00
	yes	0.00	0.00	0.00	0.00
Minimum	no	3	3	2	2
	yes	4	5	4	5
Maximum	no	4	5	5	5
	yes	4	5	4	5

Plots

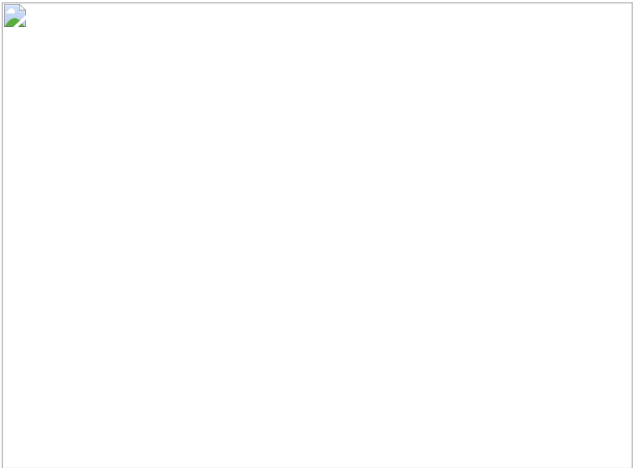
D1Exit_ArdunoConf



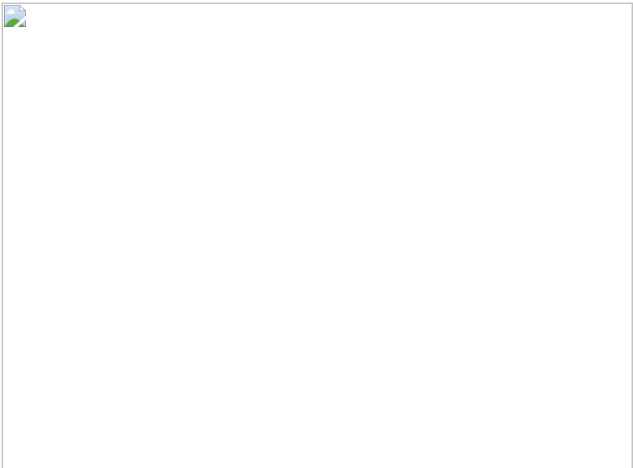
D1Exit_CircuitConf



D1Exit_troubleshootCode



D1Exit_troubleshootCircuit



Paired Samples T-Test primary outcomes

Paired Samples T-Test				Statistic	p	Effect Size
D1Entry_troubleshootCode	D1Exit_troubleshootCode	Wilcoxon W	2.50 ^a	.031	Rank biserial correlation	-0.861
D1Entry_troubleshootCircuit	D1Exit_troubleshootCircuit	Wilcoxon W	0.00 ^b	.007	Rank biserial correlation	-1.000

Note. H_a μ_{Measure 1} - Measure 2 ≠ 0

^a 2 pair(s) of values were tied

^b 1 pair(s) of values were tied

Paired Samples T-Test Processing Confidence

Paired Samples T-Test

			Statistic	p	Effect Size	
D1Entry_ProcessingConf	D2Exit_ProcessingConf	Wilcoxon W	0.00	.034	Rank biserial correlation	-1.00

Note. $H_1: \mu_{\text{Measure 1}} - \mu_{\text{Measure 2}} \neq 0$

Descriptives

	N	Mean	Median	SD	SE
D1Entry_ProcessingConf	6	1.17	1.00	0.408	0.167
D2Exit_ProcessingConf	6	3.50	3.50	0.548	0.224

References

[1] The jamovi project (2025). *jamovi*. (Version 2.7) [Computer Software]. Retrieved from <https://www.jamovi.org>.

[2] R Core Team (2025). *R: A Language and environment for statistical computing*. (Version 4.5) [Computer software]. Retrieved from <https://cran.r-project.org>. (R packages retrieved from CRAN snapshot 2025-05-25).

[3] Kerby, D. S. (2014). The simple difference formula: An approach to teaching nonparametric correlation. *Comprehensive Psychology*. [link](#), 3, 2165–2228.